Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Paul Kwiat	Email	Kwiat@illinois.edu
Institution / Department	*University of Illinois Urbana-Champaign/Depts. of Physics and Electrical and Computer Engineering*			
Address	1110 W. Green St., Urbana, IL 61801 USA			

2. Technology Overview

Title of Proposed Technology	*Quantum Photonic Timing Technology*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (X) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical - Electronics		*Electrical - Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Optical Measurement & Control Devices/Light Sources*	*Measuring Instruments*	*Optical Measuring Instruments*
Concept of Proposed Technology	Non-linear optical detection and optical quantum "entanglement" can be used to realize quantum-advantaged optical sensors. We have demonstrated a non-linear optical timing method capable of distinguishing single photons separated by less than 1 picosecond ($<10^{-12}$ sec). Moreover, we can realize timing measurements with resolutions below 10 attoseconds ($<10^{-17}$ sec) using quantum entanglement, even with photon detectors that have timing jitters a million times larger. Both of these open the door to dramatically improved LiDAR. Our non-linear detector can measure 30-micron steps, and we have already demonstrated the use of quantum entanglement to extract time-varying signals (e.g., measuring 20-nm vibrations). Both techniques also enable remote clock synchronization, a key capability for navigation and distributed systems. Together, these results demonstrate a genuine quantum advantage, e.g., resistance to loss and background, achieving performance and functionality that are fundamentally inaccessible to classical optical methods. Furthermore, because they do not intrinsically require cryogenics, they can be implemented in low Size, Weight, and Power (SWaP) packages; e.g., we have implemented similar systems on drones, and in a small platform on the International Space Station. Advancing the proposed quantum sensing solutions to the same level of robustness and form factor would enable a host of new applications.			

Rationale for Joint R&D Project	While we have successfully demonstrated compelling proof-of-principle results showing genuine quantum advantage in optical sensing, translating these capabilities into **practical, deployable technologies** requires a level of system integration and engineering that goes beyond our own expertise. Achieving real-world utility will demand **robust, scalable, and field-ready implementations** of key components, including stable laser sources, low-noise single-photon detectors, integrated waveguide-based sources of entanglement, and nonclassical interferometers that maintain performance in realistic environments. This joint R&D project is essential because these challenges sit at the **intersection of quantum optics, photonic engineering, device fabrication, and systems integration**. No single group can independently optimize all of these elements while also ensuring compatibility, reliability, and manufacturability. Close collaboration enables co-design across hardware, packaging, and control layers, accelerating progress from laboratory demonstrations to operational prototypes. Moreover, collaboration allows rapid iteration between theory, experiment, and engineering, ensuring that quantum advantages remain robust against loss, background noise, and environmental fluctuations. By combining complementary expertise and infrastructure, this partnership is uniquely positioned to transform promising quantum sensing concepts into **deployable platforms** with clear impact in navigation and remote sensing—outcomes that cannot be achieved through isolated efforts alone.
Keywords	1. Quantum-enhanced optical sensing 2. Ultra-precise timing and ranging 3. Integrated photonic entanglement sources 4. Low SWaP optical metrology

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

Single-photon detectors with low timing jitter: Time-of-flight LiDAR determines distance by emitting short optical pulses and measuring the round-trip travel time of reflected photons, with range resolution directly set by timing uncertainty. Modern single-photon LiDAR systems typically achieve millimeter- to centimeter-scale depth resolution, corresponding to timing jitters of several to tens of picoseconds (McCarthy et al., 2013). In principle, far finer resolution is possible. Ultrashort optical pulses, or time-frequency-entangled photon pairs generated via Spontaneous Parametric Down-Conversion (SPDC), where a narrowband pump photon is converted into two broadband daughter photons, can exhibit correlation features on femtosecond time scales (MacLean et al., 2018). However, this intrinsic precision is largely negated by detector timing jitter. Silicon avalanche photodiodes (APDs) typically exhibit tens to hundreds of picoseconds of jitter; and while record Superconducting Nanowire Single-Photon Detectors (SNSPDs) have demonstrated sub-3 ps timing resolution at the cost of very low detection efficiency (Korzh et al., 2020), values exceeding 50 ps remain common in commercial devices.

Is it possible to achieve better photon detection timing resolution? Yes – one can use a short escort pulse to frequency-convert the signal of interest in a non-linear material (Eckstein et al., 2011). In this case, the frequency conversion will only occur if the two pulses overlap, leading to an uncertainty given by the convolution of the two pulses. One problem with this method is that if the pulses do *not* overlap, then there is no signal, nor any information by how much the pulses are not overlapping.

To remedy this, we are developing a technique that we call **energy-of-light chronometry (ELC).** With this method, we put a strong chirp on the escort pulse, so that the front end corresponds to redder (less energetic) photons, while the back end corresponds to bluer (more energetic) photons. We still combine this in a non-linear crystal with the signal of interest. However, now the color (i.e., energy) of the resulting frequency-converted photon will depend on the precise relative temporal offset between the signal and the center of the chirped escort pulse. This is shown in Fig. 1 below. Therefore, by simply measuring the wavelength of the converted photon using a high-resolution spectrometer, we can infer the timing of the incident photon to much better than the detector jitter time. Moreover, the converted photon might be at a wavelength that is easier to detect (e.g., a 1560-nm photon, upconverted using a 780-nm chirped laser pulse, will create a photon around 520 nm, which can be efficiently detected with a standard Si APD). We can tune the dynamic range and timing resolution of this technique by altering the amount of chirp we place on the escort pulse. Importantly, the resolution of this technique depends on both the pulse duration of the signal and the amount of chirp we place on the escort. So, to extract the best timing resolution of pulses of various pulse durations requires the system to be reconfigured to supply a different amount of chirp on our escort. We currently can tune the chirp by sending our escort down various lengths of 780HP optical fiber. Ultimately, we have demonstrated that the photon timing information can be extracted with picosecond or sub-picosecond resolution, orders of magnitude smaller than the timing jitter of the photon detectors.

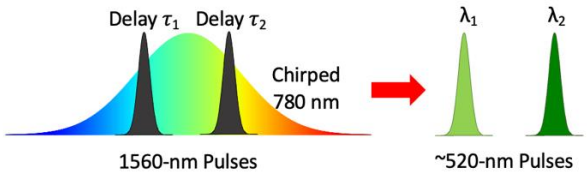

Fig 1. Illustration of the ELC protocol. A 780-nm pulse is chirped using a long fiber. The chirped pulse overlaps with 1560-nm signal photons in an up-conversion crystal to produce ~520-nm photons whose wavelengths encode their relative timing. Measuring the color of the upconverted photons with a spectrometer thus reveals timing information.

Attosecond-scale quantum interferometry: Classical single-photon interference, where light (or single photons) travel in a superposition of two paths and interferes with itself on a beamsplitter, is an effective tool for precision measurements of relative delay (whether temporal or spatial). However, the measurement accuracy can be adversely affected by optical loss or background, which can distort the interference signal. In contrast, quantum two-photon interference (Hong et al., 1987) involves two photons meeting on a beamsplitter. If these photons are temporally indistinguishable (i.e., arriving simultaneously) in addition to being identical in every other degree of freedom (e.g., color, polarization, etc.), they will "bunch up" and exit the beamsplitter in the same port. The presence (or lack) of bunching can be used to make relative delay measurements in a manner similar to classical interference. However, the quantum effect is unique in that it is inherently robust against background and loss, which has important implications for sensing applications in optically unforgiving settings. However, it is difficult for quantum interference to achieve a resolution comparable to classical interference; ultrabroadband photons or long measurement times are typically required, both of which often present practical limitations.

Is it possible to achieve better resolution with quantum interference? Yes – one can introduce color (i.e., energy) entanglement between the two photons undergoing interference. When two photons are energy-entangled, their energies are correlated in a manner that can be explained only by quantum mechanics. With **energy-entangled quantum interferometry (EEQI)**, the conventional two-photon interference signal exhibits a rapid modulation that depends on how different the two entangled energies are; the greater the difference, the faster the modulation. This modulation significantly increases the attainable temporal resolution without requiring long measurements or broadband photons; our group has thereby demonstrated attosecond-scale timing resolution with measurements that take only a matter of seconds (or faster) that also exhibit a quantum advantage over classical methods in the presence of background and loss (Lualdi et al., 2025). Figure 2 provides an overview of an EEQI measurement. Additionally, we have also begun expanding the system capabilities to measure time-varying path lengths, as might occur, e.g., due to vibrations of some distant object off of which the photons are reflected; we have already experimentally demonstrated that our system can detect vibration amplitudes as low as 20 nm, at frequencies up to 20 kHz.

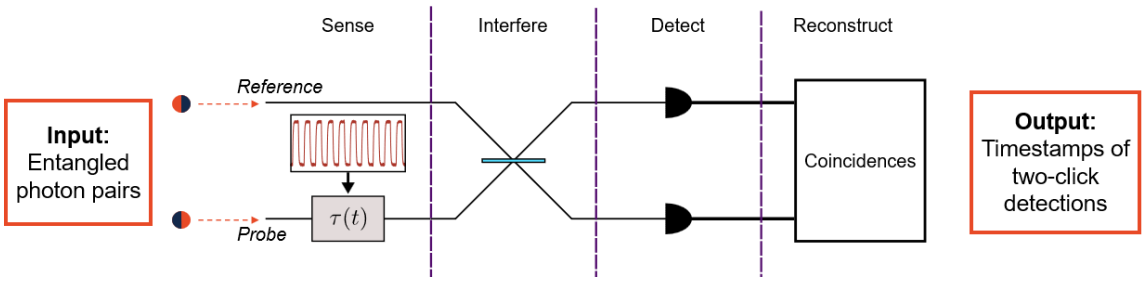

Fig 2. Illustration of an EEQI measurement. During the sense step, two energy-entangled photons—reference and probe—are incident on a beamsplitter. The probe photon experiences some relative delay, which could be time-dependent (i.e., bouncing off a vibrating surface). These photons then interfere on a beamsplitter. Detectors at the output observe for the bunching effect, which manifests as coincidence events (two-click detections). These detections allow us to reconstruct the (possibly time-varying) delay experienced by the probe photon.

Current experimental capabilities: Our laboratories house all the components necessary to demonstrate table-top ELC and EEQI demonstrations. Particularly, we have low-jitter SNSPDs (both visible and telecom-sensitive versions), APDs, and high-resolution (8-50 ps) time taggers. For ELC, we have multiple ultrafast telecom-optical clocks and higher-harmonic generating modules to produce escort pulses and fiber spools to impart varying amounts of chirp to these pulses. We also have both visible and telecom-sensitive high-resolution spectrometers to detect frequency conversion. For EEQI, we have a source that produces entangled photons with very different energies. We also have an advanced interferometer and detector system required for the "interfere" and "detect" steps in Fig. 2.

Looking ahead: While each of these methods is interesting on its own, e.g., for ultraprecise enhanced LIDAR, or remote vibration detection, these techniques can also be combined to realize critical applications such as remote clock synchronization. For clock synchronization, the Bahder protocol (Bahder et al., 2004) requires a mechanism to equalize the distances between two clocks from a central station. Photons are then sent from the central station to the remote clocks, where their detection timestamps facilitate the clock synchronization. EEQI enables robust path equalization, while ELC supports high-precision clock synchronization by detecting photons with a high temporal resolution. Finally, the range of applicability of these quantum-enhanced sensing modalities will be enormously enhanced by implementing them with sufficiently low Size, Weight, and Power (SWaP) that they can be deployed, e.g., on a moving vehicle or a drone. Although the current tabletop demonstrations would not permit this (TRL 3), one overall objective of this research is to create and evaluate a low-SWAP deployable prototype (TRL ≥6)

R&D Objectives: To implement these methods in a low-SWaP configuration, we must make three key R&D advancements, leveraging integrated-optics techniques where possible:

1. **Compact chirped quantum pulse gate for ELC:** Currently, we introduce chirp by sending light through 30 m of fiber and use a bulk-optics non-linear crystal, both of which introduce unnecessary loss and mass. Bulk-optics crystals are also far less efficient at up-conversion than their integrated-optics equivalents. **We seek to significantly reduce the SWaP by devising an integrated-optics up-conversion device and researching alternative chirping devices.** We plan to investigate two main chirping techniques. Both a Bragg reflector and a pair of diffraction gratings can introduce the required degree of chirp while providing lower weight and/or better transmission of escort power (i.e. low-SWaP). An integrated up-conversion module will enable the use of a low-power (i.e., low-SWAP) optical pump by virtue of increased efficiency (less power loss).

2. **Compact single-photon spectrometer for ELC:** We are currently measuring the spectrum of up-converted photons by leveraging a quantum-inspired amplification technique, “weak-value amplification” (Starling et. al., 2010). However, this technique intrinsically utilizes a small fraction of the upconverted light and requires the use of a sensitive interferometer with a separate set of single-photon detectors, which complicates the creation of a compact and mobile version. **We seek to leverage a single-photon avalanche detector (SPAD) array to realize a compact and non-interferometric spectrometer with single-photon sensitivity.**

3. **Compact source of tunable energy entanglement for EEQI:** EEQI requires a bright source of energy-entangled photon pairs. Furthermore, tunable entanglement enables adjustable measurement sensitivity, which benefits many protocols. Currently, we partially meet these requirements by utilizing a bulk-optics source. While bright, this source is neither compact nor tunable. Previous attempts to use entanglement sources implemented in an integrated-optics platform were limited by loss introduced by non-optimal design. **We seek to develop an integrated-optics source that directly generates the required quantum states with tunable energies.**

R&D Target Specifications

Objective 1: Compact chirped quantum pulse gate

Specification	Target value
Single-photon timing resolution	≤5 ps
Up-conversion efficiency	≥30%

Objective 2: Compact single-photon spectrometer

Specification	Target value
Wavelength resolution	≤100 picometers
System detection efficiency	≥30 %

Objective 3: Compact source of tunable energy entanglement

Specification	Target value
Center wavelength	1550 nm
Energy non-degeneracy	0 THz through ≥10 THz (tunable)
Brightness	≥10^5 pairs per second per mW of pumping power

R&D Plan

Compact chirped quantum pulse gate (Objective 1): There are several possible avenues to integrate our non-linear frequency conversion onto chip. We currently use periodically poled lithium niobate (LN) bulk crystals, and LN waveguides have already been well researched and developed by several of our research collaborators (Ma et al., 2018, Chae et al., 2022). Without changing the material or phase-matching conditions, we can leverage an already well-understood technology to quickly improve our system. At the same time, we plan to research other non-linear techniques. Silicon waveguides could provide several advantages. For example, the sum-frequency generation output power, as performed in the LN waveguide, scales linearly with pump power and therefore requires large pulse energies (~microJoules) to acquire ~60% conversion efficiency (Bock 2020, VanDevender 2007). If we instead perform a third-order, non-linear process in silicon, the output power should quadratically scale with pump power and could potentially provide high conversion efficiency at lower pulse energies, especially if implemented in waveguide devices (Du et al., 2025, Bell et al., 2016). Similarly, there are several areas of noise in LN caused by the pump (Raman scattering, other second-order processes, etc.), that are repressed in silicon, because silicon is a centrosymmetric media, and therefore does not allow even non-linearities.

At the same time, we plan to investigate several chirping techniques: a Bragg reflector can introduce the required degree of chirp over a far shorter path length and would therefore provide the best low weight technique; however, it does not provide a way to tune the amount of chirp placed on the escort pulse, a requirement when working with various pulse durations, and would require coupling the system into fiber. Alternatively, a pair of diffraction gratings with high line density (~2400 lines/mm) can provide tunable chirp by altering the distance between the gratings over a relatively small travel distance (on the order of 10s of cms). Furthermore, with gratings the chirp on the escort could be altered without reconfiguring the system and low-loss gratings could improve transmission and production of our chirped escort, lowering the power requirements of the system.

Compact single-photon spectrometer (Objective 2): Recent work has demonstrated the use of a linear single-photon avalanche detector (SPAD) array to realize a single-photon spectrometer offering 40-pm spectral resolution and 40-ps temporal resolution (Jirsa et al., 2025). Since the up-converted photon generated by our chirped up-conversion technique will have a wavelength suitable for silicon SPADs, we can leverage this technique to meet our spectroscopy requirements while satisfying low SWaP, since silicon SPADs do not require cryogenic cooling.

Jirsa et al. report “Alignment of the spectrometer optical setup with respect to the very thin linear

LinoSPAD2 array was a difficult task, requiring precision." To address this challenge, we will collaborate with our SME partner to design a monolithic optical payload containing a linear SPAD array and a diffraction grating. Our design contributions will draw on our experience with compact and robust optical payloads for mobile airborne (4) and spaceborne (5) platforms. After manufacture, we will characterize the spectral and temporal resolutions with standard spectral calibration sources. Once all specifications are deemed satisfactory, we will perform ECL measurements.

Compact source of tunable energy entanglement (Objective 3): To reduce the SWaP of our entanglement source, we will leverage prior work reported in the literature demonstrating energy-entangled pairs generated directly from a non-linear waveguide device (Eckstein et al., 2008). In this work, the observed energy difference is too small for useful sensing. We will collaborate with our SME partner to develop a similar source but modify the non-linear characteristics to instead achieve a larger energy difference (≥10 THz) with photons centered around 1550 nm. Temperature tuning will be utilized to tune the energy difference.

Prior to manufacture, we will simulate the waveguide source to determine the design parameters required to satisfy our objectives. Once we receive the source, we will utilize our existing single-photon spectroscopy tools to confirm the presence (and tunability) of the energy entanglement. We will also measure the source brightness by observing the number of entangled photons produced. Finally, we will perform EEQI measurements using photons from this source photons to verify sensing functionality.

Timeline: Key tasks and milestones

Phase I:

Months	Tasks
1-2	• Initial meetings with SME to discuss basic goals, capabilities, and principles of operation for ELC and EEQI.
3-4	• Identification of critical metrics and design "pain points". • Preliminary design of system architecture compatible with eventual low-SWaP implementation.
5-6	• Co-design of final prototypes, including 1-2 risk-reduction alternatives. • Creation of Phase II proposal.

Phase II:

Quarter	Tasks
Q1	• Acquire required new hardware, e.g., chirping grating, waveguide up-conversion device, SPAD array, waveguide entanglement device
Q2	
Q3	• Test and verify components
Q4	• Assemble and optimize subsystem evaluation testbeds (compact quantum pulse gate, compact spectrometer, and entanglement source)
Q5	
Q6	• Sub-system evaluation: Confirm performance specifications • Design full ECL and EEQI prototypes with newly developed low-SWaP components
Q7	
Q8	• Implement ECL and EEQI demonstrations
Q9	• Evaluate ECL and EEQI prototypes
Q10	• Deploy quantum sensor prototype on mobile platform • Plan for future development to higher (>6) TRL

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The UIUC researchers will collaborate closely with the Korean SME Partner during the design process, drawing on their deep experience and expertise to inform design choices to minimize SWaP as much as possible while maximizing the success probability. We will also lead the characterization of the newly developed low-SWaP components and subsystems, as we have extensive experience with the relevant characterization techniques, and have access to a full complement of quantum optics equipment to support such measurements. Finally, as experts with the ECL and EEQI techniques underpinning the proposed research program, we will lead the integration of newly developed low-SWaP components and subsystems into functioning ECL and EEQI systems. With the SME partner we will seek to identify and explore future applications.
Expected R&R of Korean SME Partner	The Korean SME partner will collaborate closely with the Research Institute during the design process. Korean SME partner team members will draw on their experience with industrial design and developing robust commercial-grade optical systems to inform design choices to minimize SWaP as much as possible while ensuring that the desired technical functionality can be achieved. The Korean SME partner will also lead the the manufacture of the novel low-SWaP components and subsystems developed during this research program. The SME partner will draw on their experience with manufacturing commercial systems, and leverage in-house manufacturing capabilities where possible to support quality control, enable design flexibility, and allow for novel designs beyond commercially available products.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	We expect to demonstrate exceptionally low-jitter photon detection and attosecond-scale temporal delay measurements when implementing our ECL and EEQI techniques with compact, low-SWaP components. Importantly, these new and low-SWaP components will allow us to, for the first time, demonstrate high-resolution timing measurements on mobile platforms. Our anticipated results will play a significant role in addressing real-world technical challenges with LiDAR, remote clock synchronization, and other applications that require high-resolution photon detection and precise measurements of relative delay. Quantum-enhanced commercial sensing systems will become a real possibility. The developed technologies could also potentially be combined with imaging capabilities to open up new sensing capabilities. For example, the high temporal resolution offered by our methods can in principle allow us to achieve much better longitudinal resolution than lateral (which would be limited by diffraction) for “reasonable” telescope apertures. This would enable enhanced precision in long-range imaging and sensing.

6. References

(1) A. Conrad et al., Drone- and Vehicle-Based Quantum Key Distribution, arXiv:2505.17587.

(2) A. Eckstein and C. Silberhorn, Broadband frequency mode entanglement in waveguided parametric downconversion, Opt. Lett., 1825 (2008).

(3) A. Eckstein, B. Brecht, and C. Silberhorn, A quantum pulse gate based on spectrally engineered sum frequency generation, Opt. Express 19, 13770 (2011).

(4) A. McCarthy, N. J. Krichel, N. R. Gemmell, X. Ren, M. G. Tanner, S. N. Dorenbos, V. Zwiller, R. H. Hadfield, and G. S. Buller, Kilometer-range, high resolution depth imaging via 1560 nm wavelength single-photon detection, Opt. Express 21, 8904 (2013).

(5) A. P. VanDevender, Quantum Information Applications of Frequency Upconversion, Ph.D. Dissertation, University of Illinois at Urbana-Champaign, 2007.

(6) B. A. Bell, J. He, C. Xiong, and B. J. Eggleton, Frequency conversion in silicon in the single photon regime, Opt. Express 24, 5235 (2016).

(7) B. Korzh et al., Demonstration of sub-3 ps temporal resolution with a superconducting nanowire single-photon detector, Nat. Photonics 14, 250 (2020).

(8) C. K. Hong, Z. Y. Ou, and L. Mandel, Measurement of subpicosecond time intervals between two photons by interference, Phys. Rev. Lett. 59, 2044 (1987).

(9) C. P. Lualdi, S. J. Johnson, M. Vayninger, K. A. Meier, S. Sahoo, S. I. Bogdanov, and P. G. Kwiat, Fast quantum interferometry at the nanometer and attosecond scales with energy-entangled photons, Science Advances 11, eadw4938 (2025).

(10) F. Ma, L.-Y. Liang, J.-P. Chen, Y. Gao, M.-Y. Zheng, X.-P. Xie, H. Liu, Q. Zhang, and J.-W. Pan, Upconversion single-photon detectors based on integrated periodically poled lithium niobate waveguides [Invited], J. Opt. Soc. Am. B 35, 2096 (2018).

(11) J. Du, X. Zhang, G. F. R. Chen, H. Gao, D. T. H. Tan, and A. Ling, Entanglement distribution over 155 km metropolitan fiber using a CMOS-compatible silicon chip, Newton 1, (2025).

(12) J. Jirsa et al., Fast data-driven spectrometer with direct measurement of time and frequency for multiple single photons, Opt. Express, 33, 9962 (2025).

(13) J.-P. W. MacLean, J. M. Donohue, and K. J. Resch, Direct Characterization of Ultrafast Energy-Time Entangled Photon Pairs, Phys. Rev. Lett. 120, 053601 (2018).

(14) J.-W. Chae, J.-H. Kim, Y.-C. Jeong, and Y.-H. Kim, Tunable up-conversion single-photon detector at telecom wavelengths, Nanophotonics 12, 495 (2023).

(15) L. M. Ramsey et al., SEAQUE: UIUC-Led Quantum Space Technology Demonstration, in Optica Quantum 2.0 Conference and Exhibition (2025), Paper QTu2A.1 (Optica Publishing Group, 2025), p. QTu2A.1.

(16) M. Bock, Polarization-Preserving Quantum Frequency Conversion for Trapped-Atom Based Quantum Networks, Doctoral Thesis, Saarländische Universitäts- und Landesbibliothek, 2020.

(17) T. B. Bahder and W. M. Golding, Clock Synchronization based on Second-Order Quantum Coherence of Entangled Photons, AIP Conference Proceedings 734, 395 (2004).